TimesFM 2.5: Multi-method forecasting comparison on BTC-USD

Automated report

April 8, 2026

1 Methodology

Zero-shot uses the pretrained TimesFM 2.5 checkpoint. LoRA loads rank-8 attention adapters trained on the same ticker. ICF (inference) uses in-context examples sampled from the training split with no leakage. Metrics are MSE/MAE on raw USD closes and directional accuracy on successive returns.

2 Summary metrics

Method	MSE	MAE	Dir. acc.
Zero-shot	66109736.395204	6049.736993	0.5148
LoRA	60802047.955078	5836.480938	0.5012
ICF (inference)	171359718.997975	10431.944445	0.4840
ICF (trained)	74265109.149195	6896.145429	0.5197

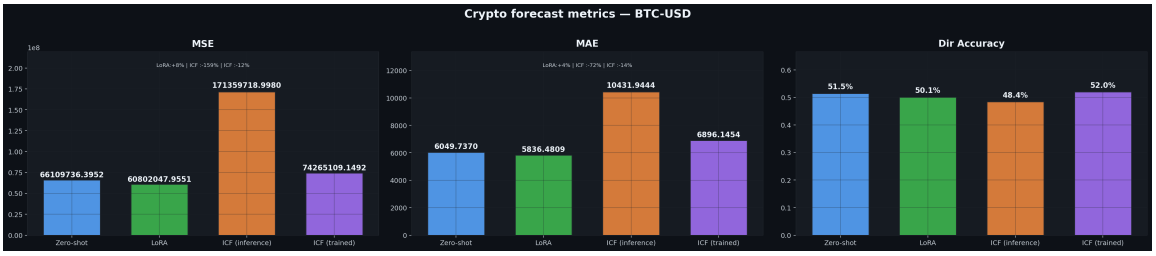
2.1 ETT (ICF benchmark)

Key	MSE	MAE	Windows
$etth1_h192$	1.0678	0.7505	105
$etth1_h336$	1.1715	0.7927	56
$etth1_h96$	0.9747	0.7102	210
$etth2_h192$	1.8801	0.9558	105
$etth2_h336$	2.2663	1.0365	56
$etth2_h96$	1.4095	0.8390	210
$ettm1_h192$	1.5763	0.8665	420
$ettm1_h336$	1.7331	0.8951	238
$ettm1_h96$	1.3031	0.7978	840
$ettm2_h192$	2.0780	0.9349	420
$ettm2_h336$	4.0987	1.0949	238
$ettm2_h96$	1.6282	0.8140	840

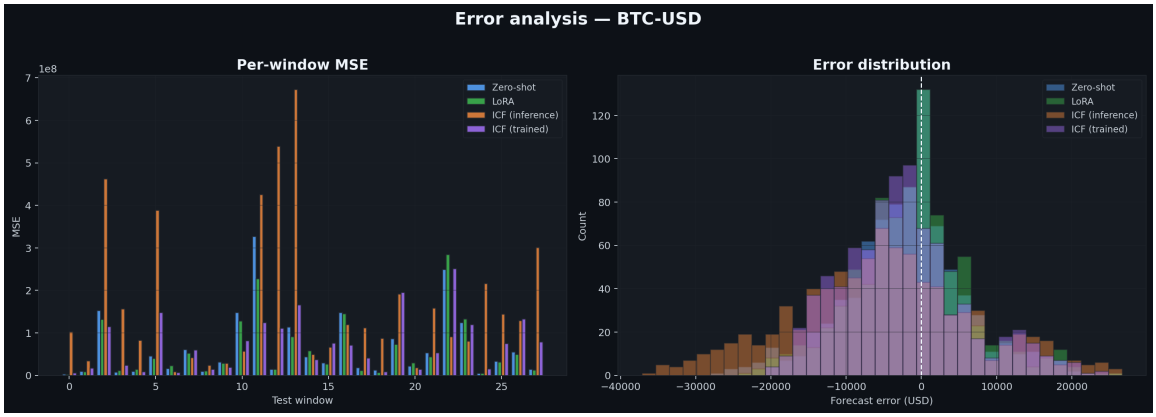
2.2 Monash (ICF benchmark, excerpt)

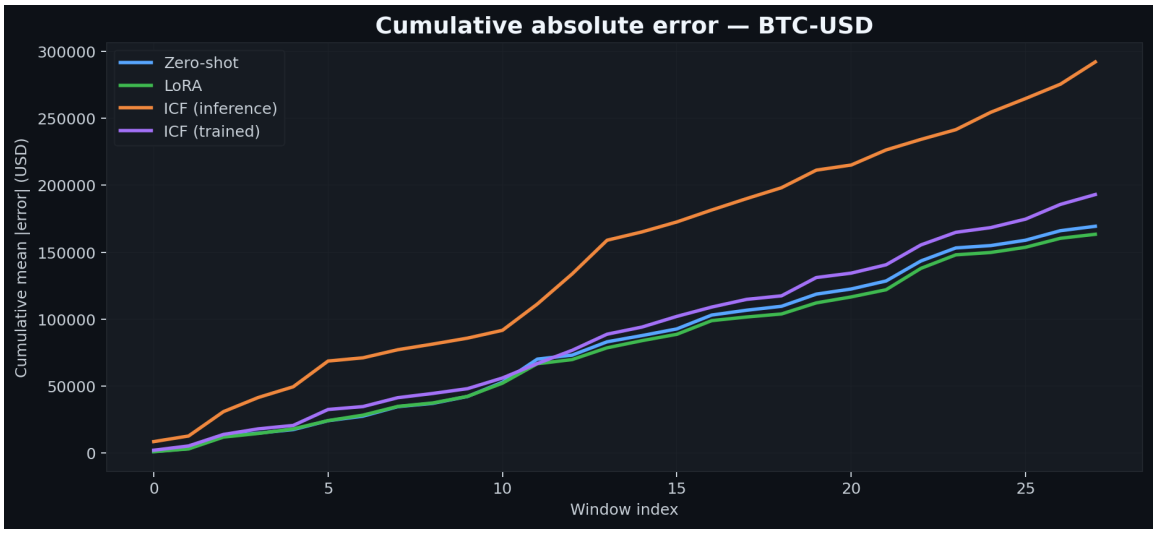
Dataset	Metric	Value
m1$_{m}onthly$	mae	8403.162445
m1$_{m}onthly$	mase	1.377211
m1$_{m}onthly$	smape	0.114375
m1$_{m}onthly$	time	2.546521
m1$_{q}uarterly$	mae	9124.806000
m1$_{q}uarterly$	mase	1.688931
m1$_{q}uarterly$	smape	0.111464
m1$_{q}uarterly$	time	1.167341
m4$_{q}uarterly$	mae	227.040738
m4$_{q}uarterly$	mase	1.539576
m4$_{q}uarterly$	smape	0.022894
m4$_{q}uarterly$	time	1.117189
m4$_{y}early$	mae	598.151197
m4$_{y}early$	mase	6.112138
m4$_{y}early$	smape	0.069176
m4$_{y}early$	time	1.121106
tourism$_{m}onthly$	mae	7056.031359
tourism$_{m}onthly$	mase	1.632731
tourism$_{m}onthly$	smape	0.078809
tourism$_{m}onthly$	time	1.118197
tourism$_{q}uarterly$	mae	16432.049699
tourism$_{q}uarterly$	mase	2.171191
tourism$_{q}uarterly$	smape	0.083297
tourism$_{q}uarterly$	time	1.094729

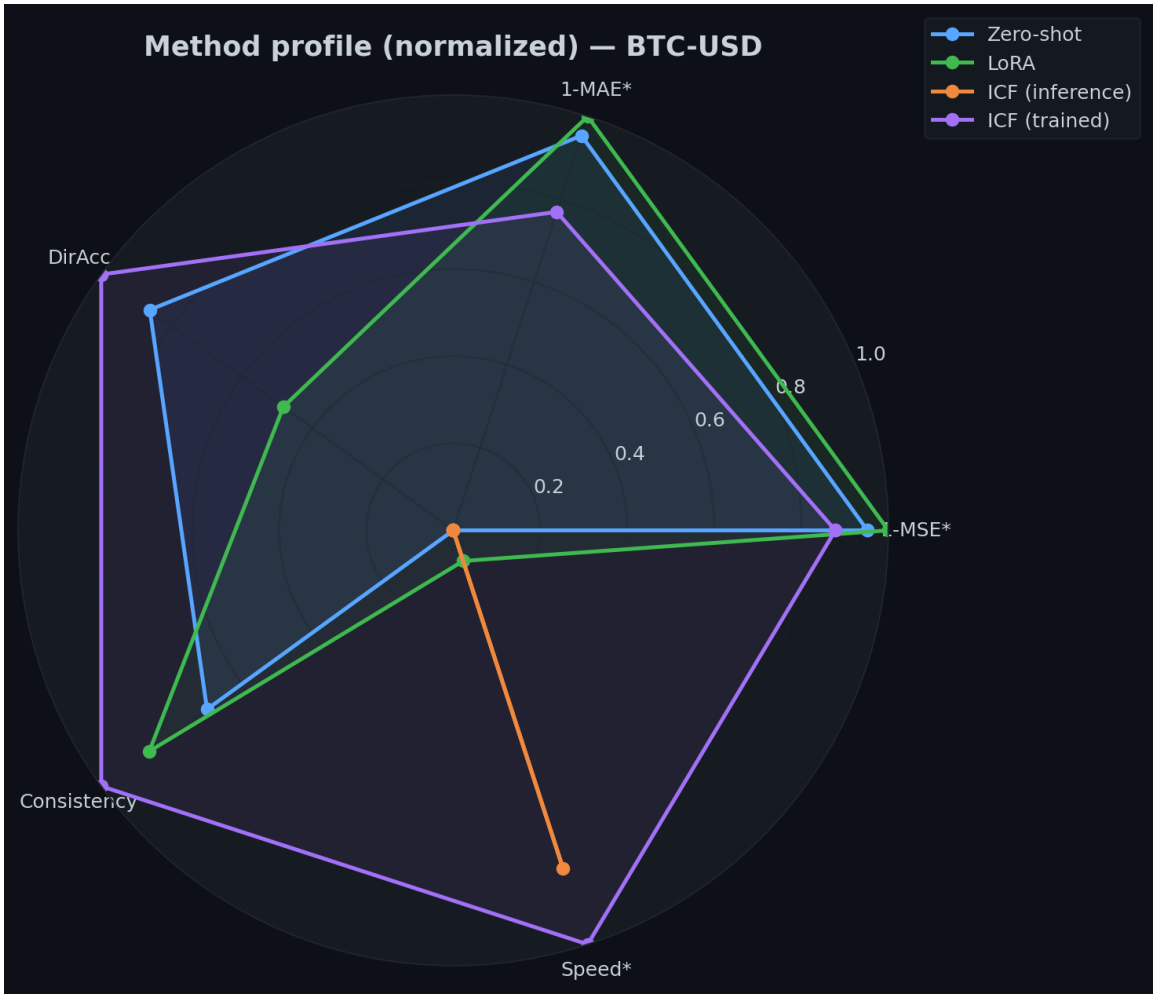
3 Figures



Forecast overlays — BTC-USD
Best ICF vs zero-shot (MSE delta) (window 23/28)
Largest MSE reduction vs zero-shot
Median zero-shot MSE window (window 26/28)
Typical difficulty
Worst ICF vs zero-shot (window 14/28)
ICF hurt most vs zero-shot
Most recent test window (window 28/28)
Context
Actual
Zero-shot
LoRA
ICF (inference)
ICF (trained)
Price (USD)
Steps relative to forecast start


Error analysis — BTC-USD
Per-window MSE
1e8
MSE
Test window
Zero-shot
LoRA
ICF (inference)
ICF (trained)
Error distribution
Count
Forecast error (USD)


Cumulative absolute error — BTC-USD
Zero-shot
LoRA
ICF (inference)
ICF (trained)
Cumulative mean |error| (USD)
Window index



4 Discussion

Lowest MSE among listed methods: lora. ICF inference without continued pretraining may underperform fine-tuning; optional ICF trained checkpoint adds a fourth curve when available.